

Neural Differential Cryptanalysis

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Outline

1. **Motivation** — Why neural cryptanalysis?
2. **Background** — Differential cryptanalysis & SPECK32/64
3. **Research Questions & Experimental Setup**
4. **Core Results**
 - ▶ Baseline · Representation · Saliency · Robustness · Markov
5. **Extended Analysis**
 - ▶ Neural vs. classical · Gen/Latent · Architecture · Efficiency
6. **Key Findings & Future Work**



Can Neural Networks Break Ciphers?

Classical differential cryptanalysis:

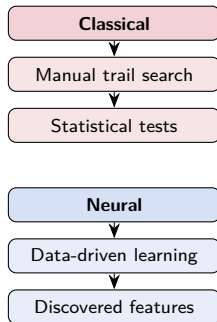
- ▶ Manual search over propagation tables
- ▶ Deep domain expertise per cipher
- ▶ Exponential search space at higher rounds

Gohr (CRYPTO 2019):

- ▶ Neural network as a *distinguisher*
- ▶ 92.4% accuracy on 5-round SPECK32/64
- ▶ Outperformed classical tests at 7–8 rounds

Open question:

Where does the signal reside, and *how* do networks extract it?





Differential Cryptanalysis & SPECK32/64

Core idea: trace input-difference propagation.

$$\Delta P = P \oplus P' \longrightarrow \Delta C = E_k(P) \oplus E_k(P')$$

Differential probability:

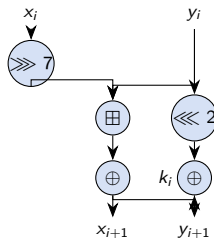
$$DP = \frac{|\{x : F(x) \oplus F(x \oplus \Delta P) = \Delta C\}|}{2^n}$$

Markov assumption: multi-round probability = product of single-round probabilities.

SPECK32/64:

ARX cipher. 32-bit block, 64-bit key, 22 rounds.

$\Delta P = 0x0040_0000$. Rounds 3–9.



SPECK32 round function



Neural Distinguishers

Reframe distinguishing as **binary classification**:

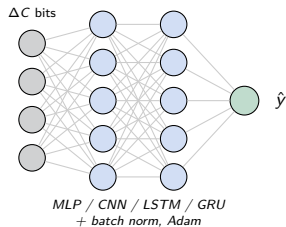
$$D_{\theta} : \{0, 1\}^n \times \{0, 1\}^n \rightarrow [0, 1]$$

- ▶ $y=1$: real cipher pair (C, C')
- ▶ $y=0$: uniformly random pair
- ▶ Loss: binary cross-entropy
- ▶ 50% accuracy = no signal

Advantage:

$$\text{Adv} = |\Pr[D=1|y=1] - \Pr[D=1|y=0]|$$

Learns non-linear decision boundary from data —
sidesteps sparse-sample DP.





Research Questions & Experimental Setup

RQ1 — Representation:

How does input preprocessing affect accuracy?

RQ2 — Markov Assumption:

Does the classical memoryless assumption hold?

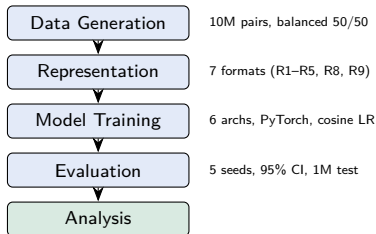
RQ3 — Architecture:

Best accuracy-efficiency tradeoff?

Setup:

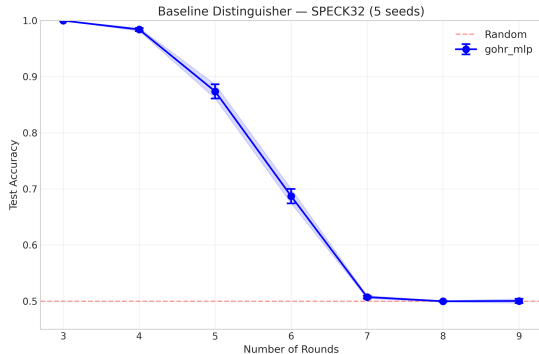
- ▶ 15 experiments on SPECK32/64
- ▶ 7 input representations (R1–R5, R8, R9)
- ▶ 6 architectures
- ▶ 10M train / 1M test per config
- ▶ 5 seeds, bootstrap 95% CIs

pp = percentage points (absolute gap).





Baseline: The Distinguishing Cliff

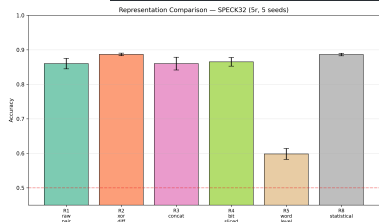


Rounds	Accuracy
3	99.99%
4	98.42%
5	87.38%
6	68.72%
7	50.73%

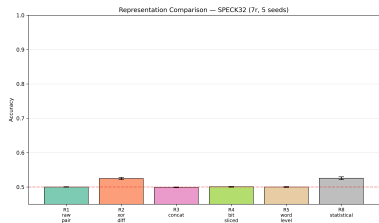
Near-perfect at 3–4, random at 7+. Sharp “cliff” between rounds 5–7. Our 87.4% at R5 vs. Gohr’s 92.4% (simpler architecture).



Representation Dominates Architecture



5 rounds



7 rounds

At 5 rounds:

- ▶ R2 (XOR diff): **88.65% $\pm$ 0.37**
- ▶ R8 (Statistical): 88.62% $\pm$ 0.36
- ▶ R5 (Word-level): 59.80% $\pm$ 1.60
- ▶ Representation gap: **28.85pp**

At 7 rounds:

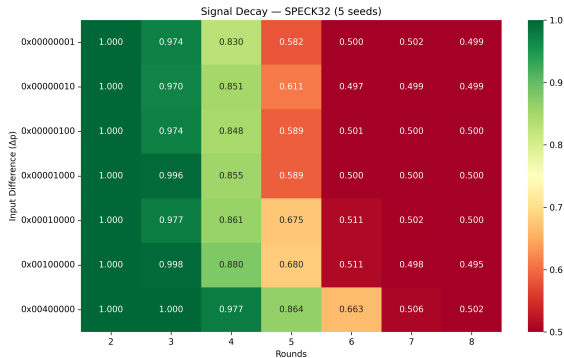
- ▶ Only R2 (52.5%) and R8 (52.6%) retain signal
- ▶ Everything else collapses to 50%

Architecture spread: only **3.89pp** (86.1–90.0%)

Takeaway: *How you preprocess matters far more than which model you use. XOR diff halves dimensionality at no cost.*



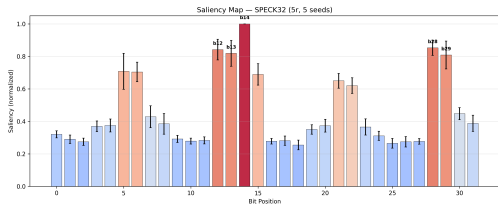
Signal Decay Heatmap



7 input differences $\times$ 7 round counts. $\Delta P=0x0040_0000$: 86.4% at R5 vs. 58.2% for 0x01. All $\rightarrow$ 50% by round 7.



The Network Discovers Cipher Structure



Gradient saliency across 32 bit positions (5 rounds)

Top-5 salient bits: 14, 28, 12, 13, 29

- ▶ 16-bit periodicity mirrors SPECK's two-word structure
- ▶ Bit 14 $\ggg$ 7 = bit 7 $\rightarrow$ modular addition carry chain

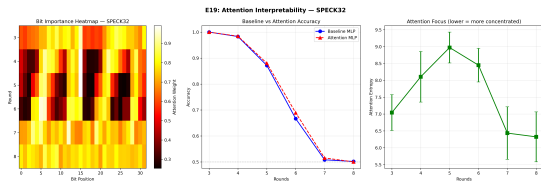
Invariance test:

Random bit permutation drops accuracy from 87.1% $\rightarrow$ **50.0%** ($-37.1pp$).

The model learns *position-specific* features tied to $\alpha=7$, $\beta=2$ — not generic Hamming weight.



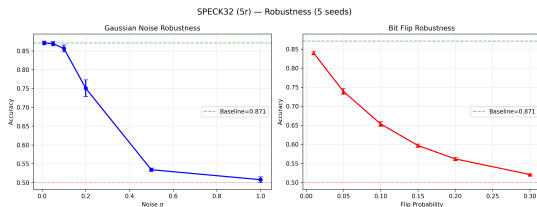
Attention Interpretation



Attention entropy decreases with round count — the model *focuses* on fewer bits as the task gets harder.
Aligns with saliency findings.



Robustness Under Noise

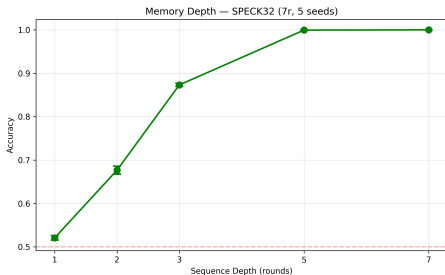


Gaussian		Bit Flip	
σ	Acc	p	Acc
0.05	86.9%	0.01	84.0%
0.10	85.6%	0.05	73.9%
0.20	75.1%	0.10	65.3%
0.50	53.4%	0.20	56.2%

- ▶ Gaussian: tolerant up to $\sigma=0.1$
- ▶ Bit flips far more destructive — even 1% $\Rightarrow -3.1pp$
- ▶ **Practical limit:** BER < 5%

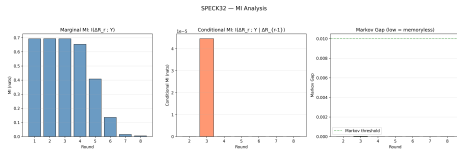


Markov Assumption: Validated



Depth analysis (7-round target)

$d=1$: 52% $d=3$: 87% $d=7$: 100%



Conditional MI ≈ 0 at all
round transitions — no residual dependency.

Three independent methods agree: depth analysis, MINE estimation, and generative modeling all confirm the Markov assumption holds for SPECK32.



Latent Space & Generative Markov Validation



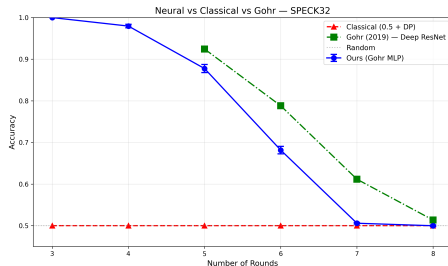
Variational Autoencoder (VAE) Latent Space:

Reveals well-separated, clustered representations corresponding directly to the underlying cipher state (real vs. random).

Validates that generated ciphertext differences maintain the exact structural properties of true differentials.

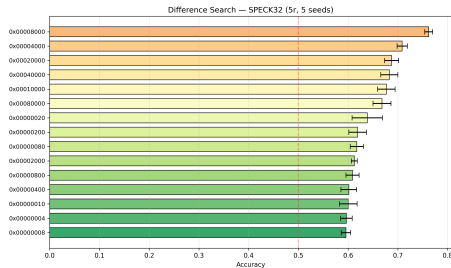


Neural vs. Classical & Automated Search



Classical sample-based DP: $< 2^{-16}$
(undetectable from finite samples).

Neural: 87.8% at R5, 68.2% at R6.



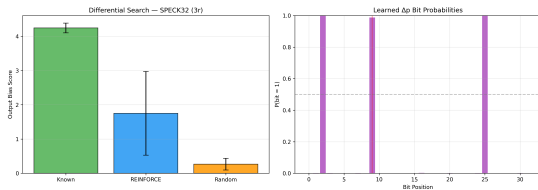
Best single-bit ΔP : 0x8000 at 76.2%.

Worst: 0x01 at 57.8%.

Top diffs cluster in bits 14–19.



RL-Based Differential Search

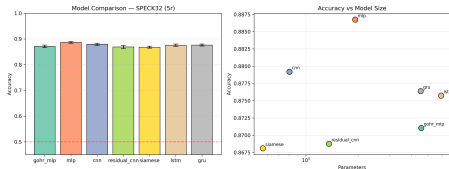


At 3 rounds: RL discovers 0x0010_2000 at **84.96%** (known-optimal: 86.63%).

At 5 rounds: insufficient gradient signal for RL exploration.

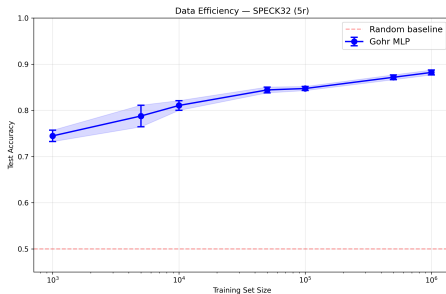


Architecture & Data Efficiency



Model	Params	Acc
LSTM	594K	90.0%
MLP	191K	88.5%
CNN	81K	88.4%
Gohr MLP	457K	86.1%

CNN: $7.3\times$ smaller, only -1.6pp .



1K samples $\Rightarrow$ 74.5%
(24.5pp above random)

Practical knee at $\sim 50\text{--}100\text{K}$.
500K $\rightarrow$ 1M: only +1.1pp.



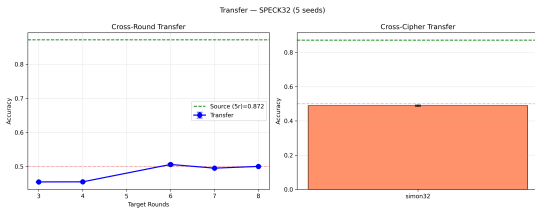
No Cross-Round Transfer

5-round model, zero-shot on others:

Target	Accuracy
3 rounds	45.5%
4 rounds	45.5%
5 (<i>source</i>)	87.2%
6 rounds	50.6%
7 rounds	49.5%

Below 50% at 3–4 = *worse than random*.

The 5-round features are anti-correlated with 3–4 round structure.





Key Findings

- ✓ **Representation > Architecture**

Representation gap: 28.85pp vs. architecture gap: 3.89pp.

XOR diff halves dimensionality with no accuracy loss.

- ✓ **Markov Assumption Holds**

Three independent validations: conditional MI ≈ 0 , generative ratio ≈ 1.0 , depth analysis consistent.

- ✓ **Position-Specific Features Tied to Cipher Structure**

37pp drop under permutation. Saliency identifies bits 14, 28, 12, 13, 29 — matching SPECK's $\alpha=7$, $\beta=2$. Automated search confirms.



Contributions & Future Directions

What we built:

- ▶ 15 experiments, 7 representations, 6 architectures
- ▶ 5-seed averaging + bootstrap CIs
- ▶ 46 unit tests, open-source PyTorch
- ▶ First systematic representation study for neural distinguishers
- ▶ Empirical Markov validation via 3 independent methods

What's next:

- ▶ Full Bayesian key-recovery attack
- ▶ Cross-cipher: SIMON32, PRESENT
- ▶ Close the 7-round gap (50.7% vs. Gohr's 61.2%)
- ▶ Neural round inversion for trail reconstruction
- ▶ Adversarial robustness beyond 5% BER

References: Gohr (CRYPTO '19) · Benamira et al. (EUROCRYPT '21) · Biham & Shamir ('91) · Belghazi et al. (ICML '18) · Beaulieu et al. ('15)

Thank You!

Questions?

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`github.com/ANGADJEET/Neural-Cryptanalysis`